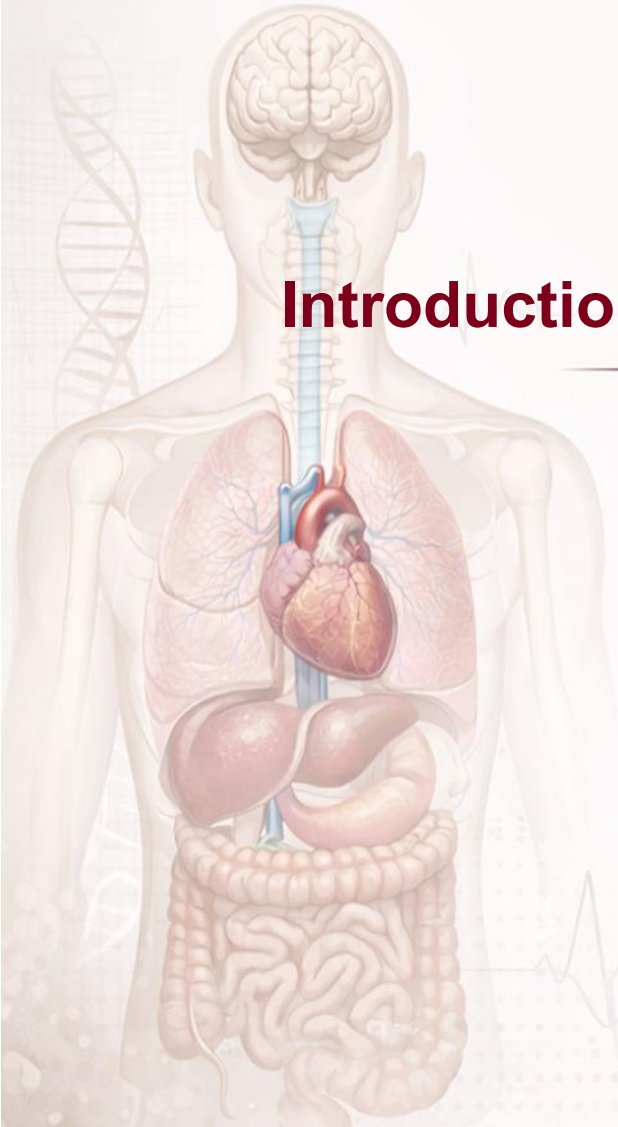


Chapter 1.
Introduction to Physiology and Homeostasis

KOREA UNIVERSITY
1905

Human Physiology (PHAR272)

전공역량 (능력)	권장이수 학년/학기					
	1학년		2학년		3학년	
	1학기	2학기	1학기	2학기	1학기	2학기
약학전문지식 이해능력	생물학의기초I 화학의기초I	생물학의기초II 화학의기초II 약사윤리	약품생화학기초 약품미생물학기초 약학실험I (미생물학/생화학) 해부생리학	약품생화학심화 약품미생물학심화 인체생리학 기본약학통계및연습	의약분자생물학 약물동태학 병태생리학I	병태생리학II 의약세포생물학
신약개발 및 첨단보건 의료 연구능력			의약품분석학I 약품유기화학 I	의약품분석학II 약품유기화학 II 신약개발개론 약학실험II(약품유기화학/분석학)	약리학 I 의약품합성학I 약학실험 III (생약학/약리학) 생약학총론	약리학 II 의약품합성학II 약학연구개론 생약학각론 약학실험IV (물리약학/약제학)
보건의료 서비스 제공을 위한 실무능력			임상약료개론	의약정보학	비처방의약품학	위장근/근골격질환임상약료
약학분야글로벌리더로서의 국제 경쟁력	1학년세미나	창업과진로	물리약학총론 천연물의약품학	물리약학각론	화장품학 의약학빅데이터	약제학총론 기능성식품학

Course type: required major course
Official language: English

Human Physiology (PHAR272)

전공역량 (능력)	권장이수 학년/학기					
	4학년		5학년		6학년	
	1학기	2학기	1학기	2학기	1학기	2학기
약학전문지식이해능력	면역학개론	면역질환과 면역제제	독성학 동물의약품학			약학세미나
신약개발 및 첨단보건의료 연구능력	예방약학I 의약화학I 약학실험V(예방약학/독성학/면역학/병태생리) 약학연구I	예방약학II 의약화학II 무기및방사성의약품학 약물타켓학 약학연구II	약물역학 바이오의약품학 AI약물설계 약학연구 III		심화연구약학실습 심화생명약학실습 심화약물과학실습 심화산업약학실습 의약품유전체학 융합약학II	융합약학II 약학캡스톤디자인 의약학중개연구
보건의료 서비스 제공을 위한 실무능력	보건사회약학 심혈관질환임상약료	내분비/종양질환임상약료 의약품규제과학 신장/정신질환임상약료	기초실무실습 감염/영양질환임상약료 호흡기/신경질환임상약료 병원약학	의료기관임상약료실습 의료기관임상약무실습 의료기관임상약제실습 지역약국약료실습 지역약국약무실습 약무행정실습 약무현장인턴쉽I	심화임상/사회약학실습 약물경제학입문 약무현장인턴쉽II	약사법규 한약제제학
약학분야글로벌리더로서의국제경쟁력	약제학각론 의약품제조관리학 및 산업약학 약업경영개론	의약품품질과학 생물약학정보학		제약생산실습 제약관리실습	창업과기업가정신	

Course type: required major course
Official language: English

Human Physiology – course plan

▷ 평가방법

항목	비율	항목	비율
출석	10%	과제	10%
중간고사	35%	기말고사	35%
수시평가	%	태도	%
참여도	10%	기타	%
합계	100%	평가점수 공개여부	공개(O) 비공개( )
평가방법 설명	- 중간/기말시험은 객관식, 단답형, 서술형 혼합 문제로 출제 - 지각 1회는 결석 1/3회와 동일하게 출석 점수 반영 - 수업 시간 중 무단이탈 시 참여도 점수 반영 - 합계 점수 (출석/과제/중간/기말/참여도)가 30점 미만 시 F 학점 부여, 참여도 점수 0점 또는 중간/기말시험 부정행위 시 F 학점 부여 - 부득이한 사정으로 시험 응시가 불가능한 경우 레포트를 제출하고 중간/기말고사 대면시험의 잔여 성적을 반영		
출석관련 규정	시험기간을 포함한 총 수업일수 1/3 이상 결석하는 경우 성적을 부여할 수 없음, 단 담당교수가 불가피한 결석으로 인정하는 경우에는 예외적용이 가능함		

▷ 강의개요

개요	본 강의는 인체의 기능을 작동하는 기작과 생체를 유지하는 방법을 소개함 또한 인체의 해부학적 측면을 탐구하여 구조와 기능 간의 연관성은 습득하고 외부환경에 대한 항상성을 유지하는 방법을 공부함
수강(권장)대상 학생	약학과 2학년
선이수과목(권장)	
수업운영원칙	영어강의
기존개설대비 변경, 개선사항	

▷ 수업목표

수업목표 일반	Use basic principles to describe the functioning of life processes associated with the topics in this course
세부목표 1	Diagram and identify the regulated homeostatic variable, sensor, integrator and effector in the homeostatically-regulated system and predict how a perturbation to the system will be compensated for
세부목표 2	Demonstrate an understanding of the physiology and basic regulatory concepts relating to the functioning of life processes

Grading: extra tests (10), attendance (10), participation (10), mid-term (35), final (35)
Planned schedules: 10/8, 15, 11/5 (conference -> online lectures to be posted)
Exams: mid-term (10/19 or 26), final (12/14 or 21), 90 min, answer in English

Human Physiology – course plan

▷ 주차별 강의계획

주	수업내용	수업핵심질문 (핵심개념)	수업교재 및 자료	과제	활동 및 설계내용
1	Introduction to Physiology and Homeostasis		1		Lecture
2	Cell Physiology & The Plasma Membrane and Membrane Potential		2, 3		Lecture
3	Principles of Neural and Hormonal Communication		4		Lecture
4	The Central Nervous System I		5		Lecture
5	The Central Nervous System II		5		Lecture
6	The Peripheral Nervous System		6, 7		Lecture
7	Muscle Physiology		8		Lecture
8	중간고사				
9	Cardiac Physiology		9		Lecture
10	The Blood Vessels and Blood		10, 11		Lecture
11	The Respiratory System		13		Lecture
12	The Urinary System		14		Lecture
13	The Digestive System		16		Lecture
14	The Endocrine System		18, 19		Lecture
15	The Reproductive System		20		Lecture
16	기말고사				

1학기는 16주 구성, 단 법정수업일수는 15주로 1주 2회 수업은 30회, 1주 1회 수업은 15회 이상 수업이 이뤄져야 하며 중간/기말고사 또는 과제물 부여는 1회 수업에 해당함 교원재량 및 학생협의에 따라, 남은 법정수업일수 외 16주차 해당하는 수업은 보강 또는 학습주로 운용 가능함

Human Physiology textbook – course plan

▷ *교재, 참고문헌 및 참고자료(동영상 및 온라인자료 포함)

Human Physiology: From Cells to Systems (9e; Cengage) - Lauralee Sherwood

1. Introduction to Physiology and Homeostasis.
2. Cell Physiology.
3. The Plasma Membrane and Membrane Potential.
4. Principles of Neural and Hormonal Communication.
5. The Central Nervous System.
6. The Peripheral Nervous System: Afferent Division; Special Senses.
7. The Peripheral Nervous System: Efferent Division.
8. Muscle Physiology.
9. Cardiac Physiology.
10. The Blood Vessels and Blood Pressure.
11. The Blood.
12. Body Defenses.
13. The Respiratory System.
14. The Urinary System.
15. Fluid and Acid-Base Balance.
16. The Digestive System.
17. Energy Balance and Temperature Regulation.
18. Principles of Endocrinology; The Central Endocrine Glands.
19. The Peripheral Endocrine Glands.
20. The Reproductive System.

1.1 Introduction to Physiology

Basic principle of human physiology



https://www.youtube.com/watch?v=LgNB3a2AltU

Definition of homeostasis

- **Homeostasis:** *self-regulating process by which biological systems maintain internal stability while adjusting to changing external conditions*



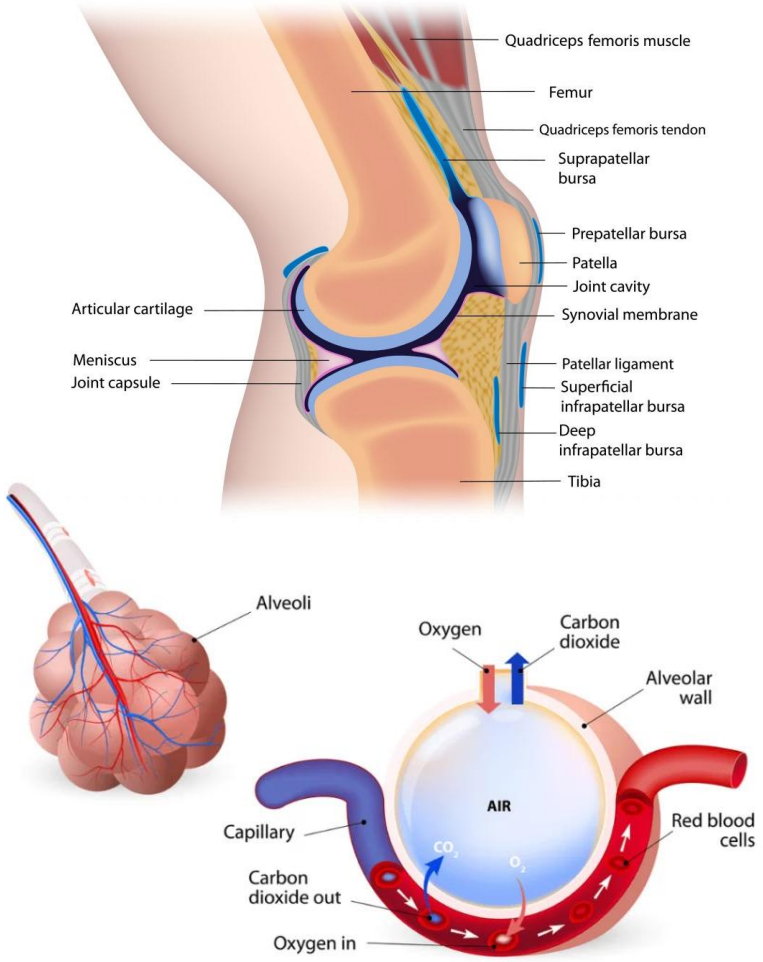
Oliver Eltinger/Agefotostock

Introduction to physiology

- Physiology is the study of the functions of living things
  - Focuses on the underlying mechanisms of body processes
- Physiology is closely related to anatomy, the study of the structure of the body
  - Structure and function are inseparable

Structure and function are inseparable





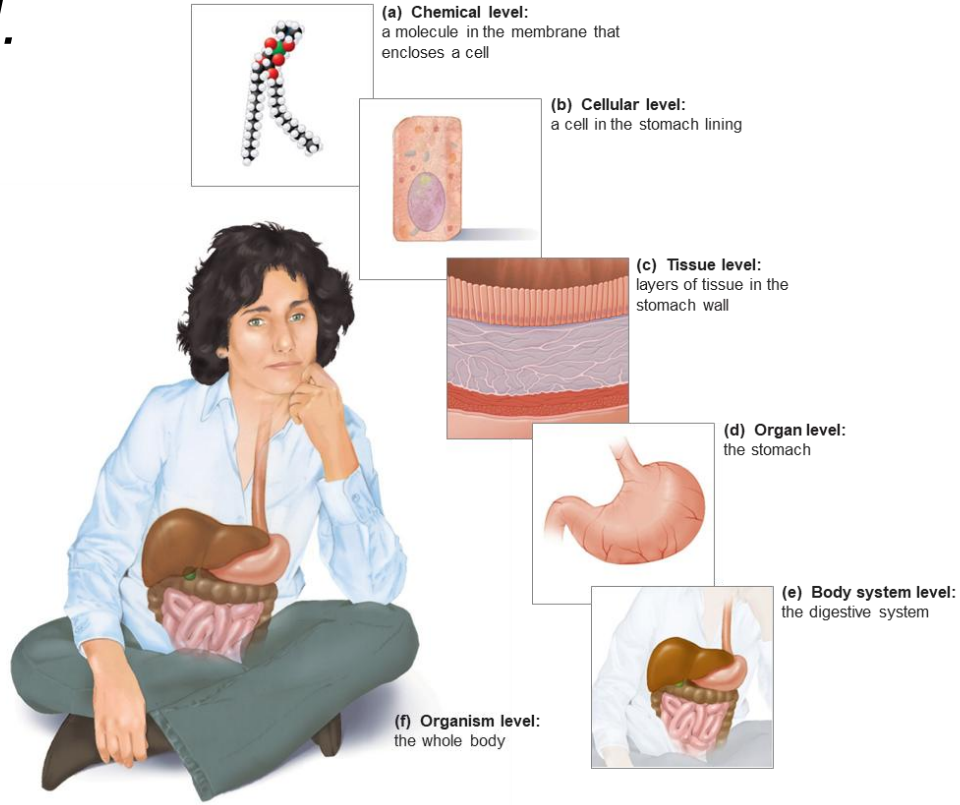
https://lowcountryfamilydentistry.com/what-is-tooth-enamel/
https://www.sapnamed.com/blog/how-to-protect-your-knees-and-make-them-healthy/
https://microfluidics-innovation-center.com/reviews/air-liquid-interface-optimized-cell-culture-substrate-microfluidic-lung-chip-application/

1.2 Levels of organization in the body

Levels of organization in the body

- **How is the body structurally organized into a total functional unit?**

The body is structurally organized into a total functional unit: At the chemical, cellular, tissue, organ, body system, and organism level.



Levels of organization in the body

- **The chemical level**

Various atoms and molecules make up the body.

- Atoms: mostly oxygen, carbon, hydrogen, and nitrogen
- Molecules of life: biomacromolecules like proteins, carbohydrates, fats, and nucleic acids (genetic material, such as deoxyribonucleic acid, or DNA)

- **The cellular level**

Cells are the basic units of life.

- Cells have basic and specialized functions (differentiation into 200 specialized types)
- Cells are progressively organized into tissues, organs, body systems, and finally the whole body

Basic cell functions – universal for every cells

- Obtaining food and O_2
- Performing energy-generating chemical reactions
- Eliminating wastes
- Synthesizing proteins and cell components
- Moving materials throughout the cell
- Responding to the environment
- Reproducing, etc

Specialized cell functions – unique for each cell type

- Gland cells of the digestive system secrete digestive enzymes that break down ingested food
- A sort of kidney cells selectively retain the required substances while eliminating unnecessary substances in the urine
- Muscle cells enable muscle contraction by producing intracellular movement
- Nerve cells generate and transmit electrical impulses to other body parts that deliver information about the external environment

Basic cell functions are essential for survival of individual cells, while specialized cell functions and intra-cellular communications are essential for the survival of the whole body

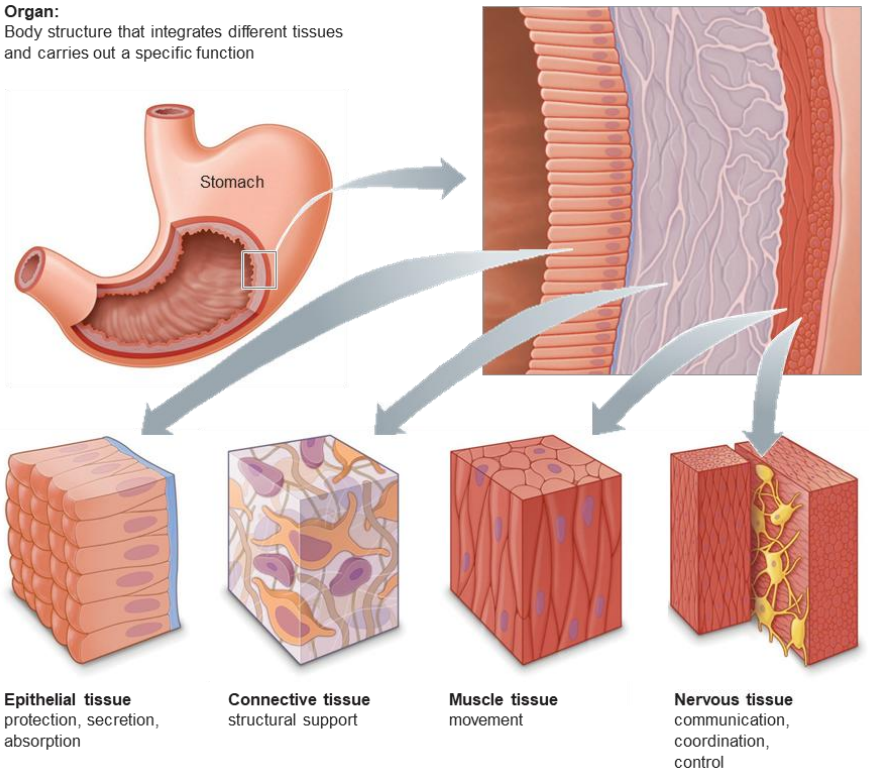
Levels of organization in the body

- **The tissue level**

Tissues are groups of cells with a similar structure and specialization.

Four primary tissues

- Epithelial tissue: barrier used for exchange of materials between the cell and environment
- Connective tissue: connects, supports, and anchors various body parts / extensive ECM
- Muscle tissue: tissue with cells specialized for contracting -> movement
- Nervous tissue: initiate and transmit electrical impulses -> communication

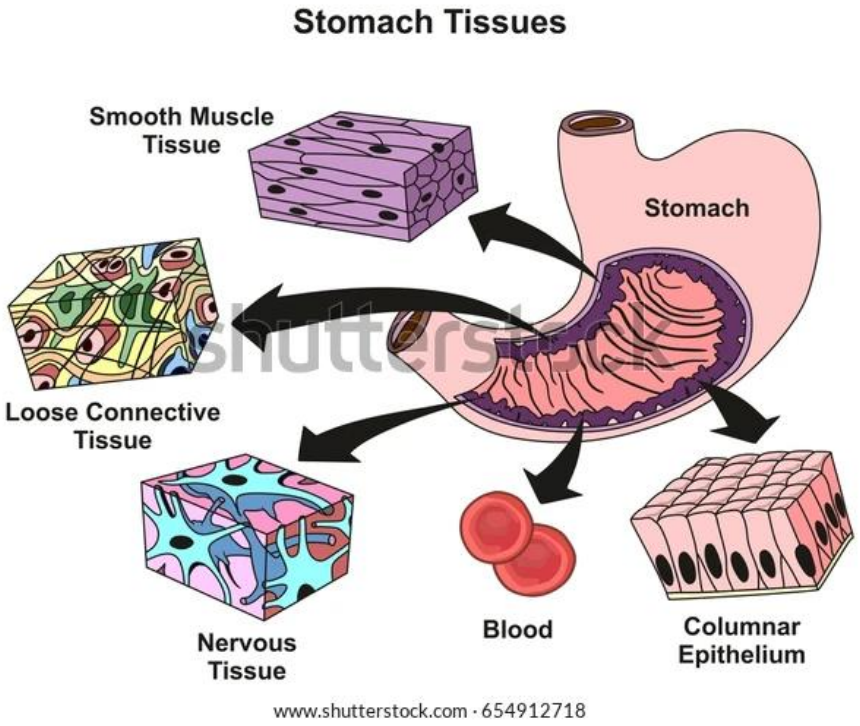


Levels of organization in the body

- **The organ level**

An organ is a unit made up of several tissue types.

- Consists of two or more types of primary tissue organized to perform particular functions
- Example: stomach, heart, blood vessels, brain, lungs, liver, kidneys, intestines, skin, pancreas, joint, bone, etc



https://www.shutterstock.com/ko/image-vector/stomach-tissues-types-structure-infographic-diagram-654912718?

Levels of organization in the body

- **The organ level**

Tissues are groups of cells with a similar structure and specialization.

An organ is a unit made up of several tissue types.

Tissue vs Organ

- Tissues: groups of similar cells working together to perform a specific function (structural building blocks)
- Organs: higher-level structures composed of two or more different tissue types, working together to perform complex, specialized body functions. (functional units like the heart or stomach)

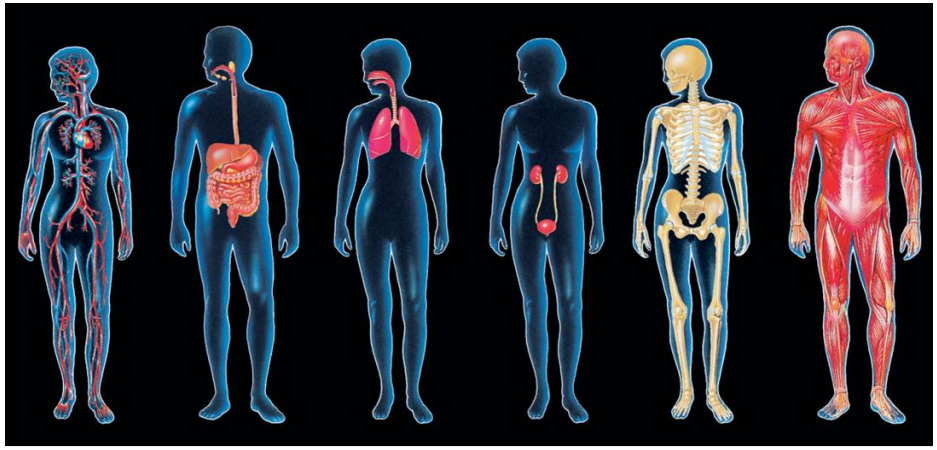
Levels of organization in the body

- **The body system level**

Body system is collection of organs performing related functions.

- Organs interact to accomplish a common activity essential for survival
- Do not act in isolation from one another

Human body has 11 body systems



Circulatory system
heart, blood vessels, blood

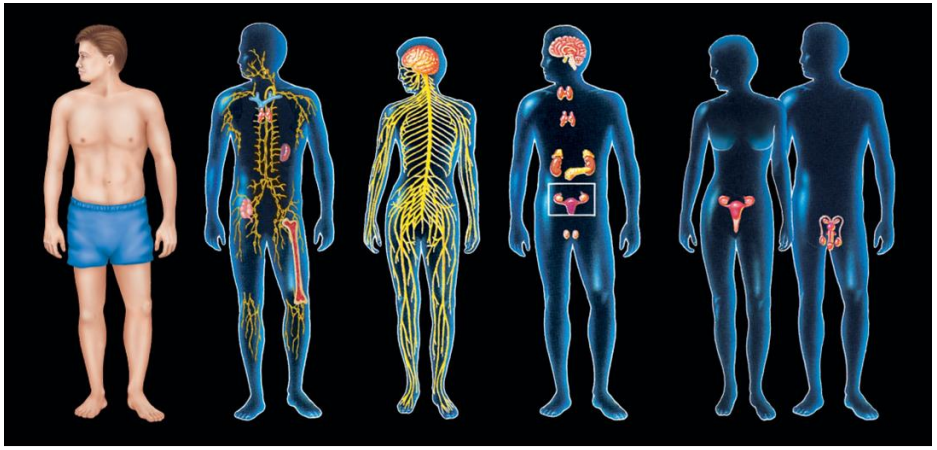
Digestive system
mouth, pharynx, esophagus, stomach, small intestine, large intestine, salivary glands, exocrine pancreas, liver, gallbladder

Respiratory system
nose, pharynx, larynx, trachea, bronchi, lungs

Urinary system
kidneys, ureters, urinary bladder, urethra

Skeletal system
bones, cartilage, joints

Muscular system
skeletal muscles



Integumentary system
skin, hair, nails

Immune system
lymph nodes, thymus, bone marrow, tonsils, adenoids, spleen, appendix, and, not shown, white blood cells, gut-associated lymphoid tissue, skin-associated lymphoid tissue

Nervous system
brain, spinal cord, peripheral nerves, and, not shown, special sense organs

Endocrine system
all hormone-secreting tissues, including hypothalamus, pituitary, thyroid, adrenals, endocrine pancreas, gonads, kidneys, pineal, thymus, and, not shown, parathyroids, intestine, heart, skin, adipose tissue

Reproductive system
Male: testes, penis, prostate gland, seminal vesicles, bulbourethral glands, associated ducts

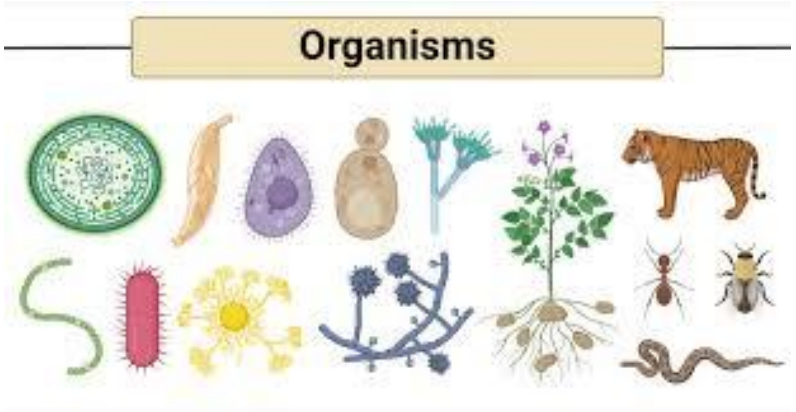
Female: ovaries, oviducts, uterus, vagina, breasts

Levels of organization in the body

- **The organism level**

Body systems are packaged into a functional whole body. An organism is any individual living entity, capable of growth, reproduction, metabolism, and responding to stimuli.

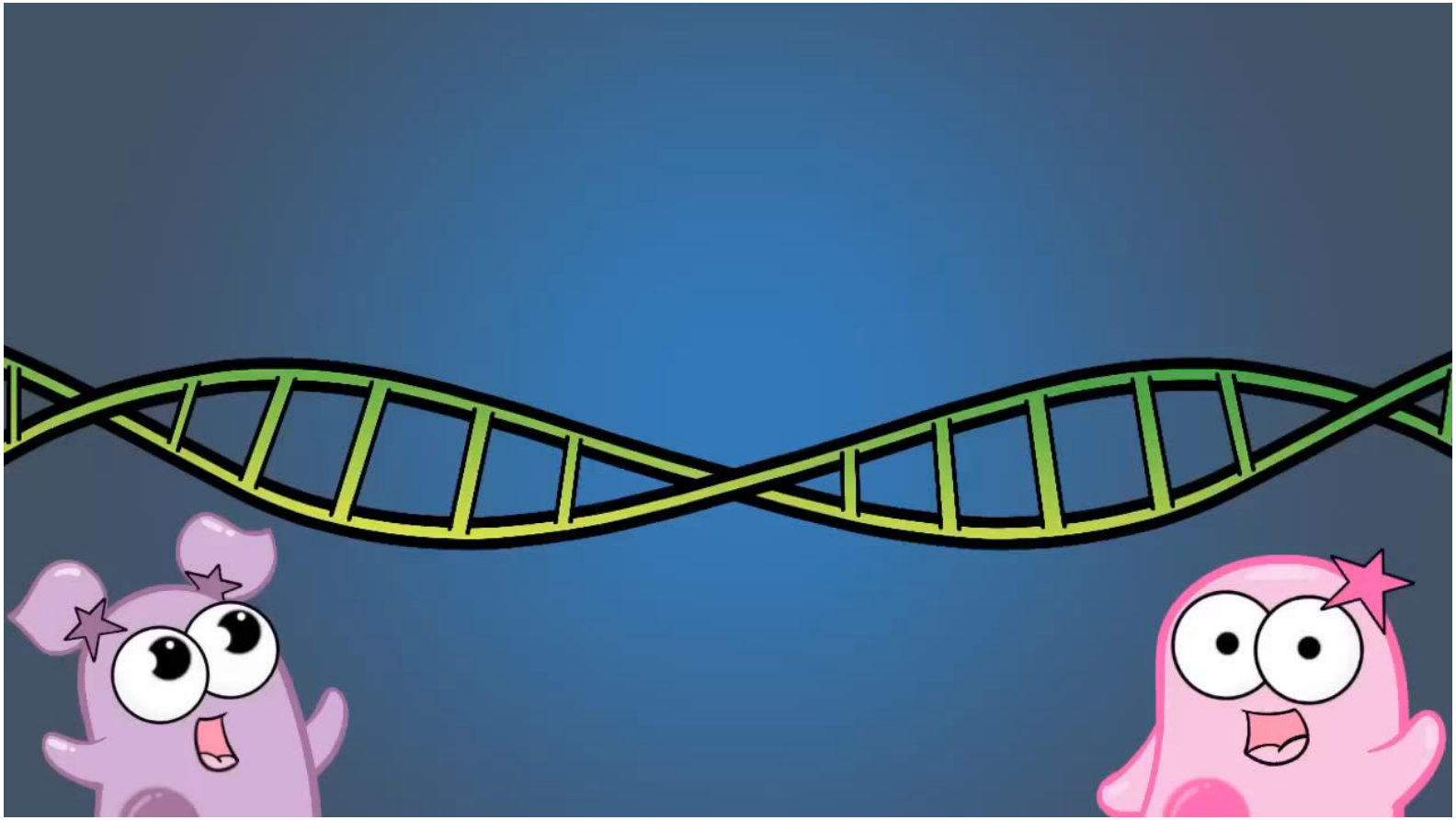
- Each body system depends on the proper functioning of other systems
- Many complex body processes depend on the interplay among multiple systems



https://microbenotes.com/organisms-classification/

1.3 Homeostasis

Homeostasis and feedback mechanism



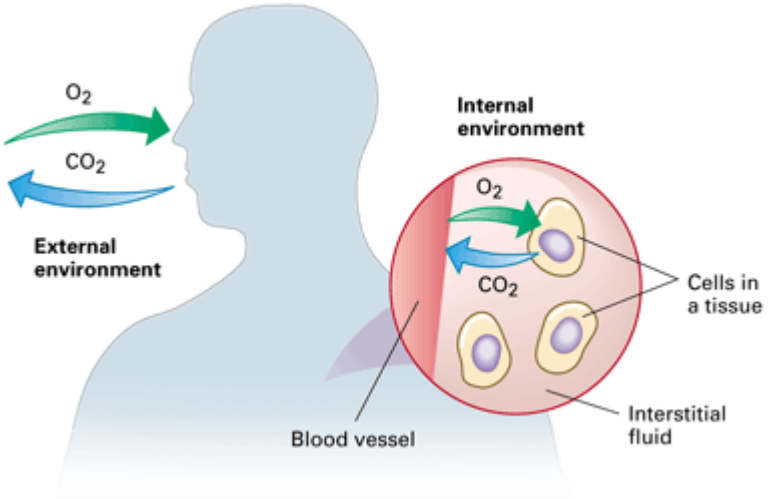
https://www.youtube.com/watch?v=Iz0Q9nTZCw4&t=30s

Concept of homeostasis

- **What is homeostasis?**

Homeostasis is a maintenance of stable internal environment against the surrounding environment where organism lives.

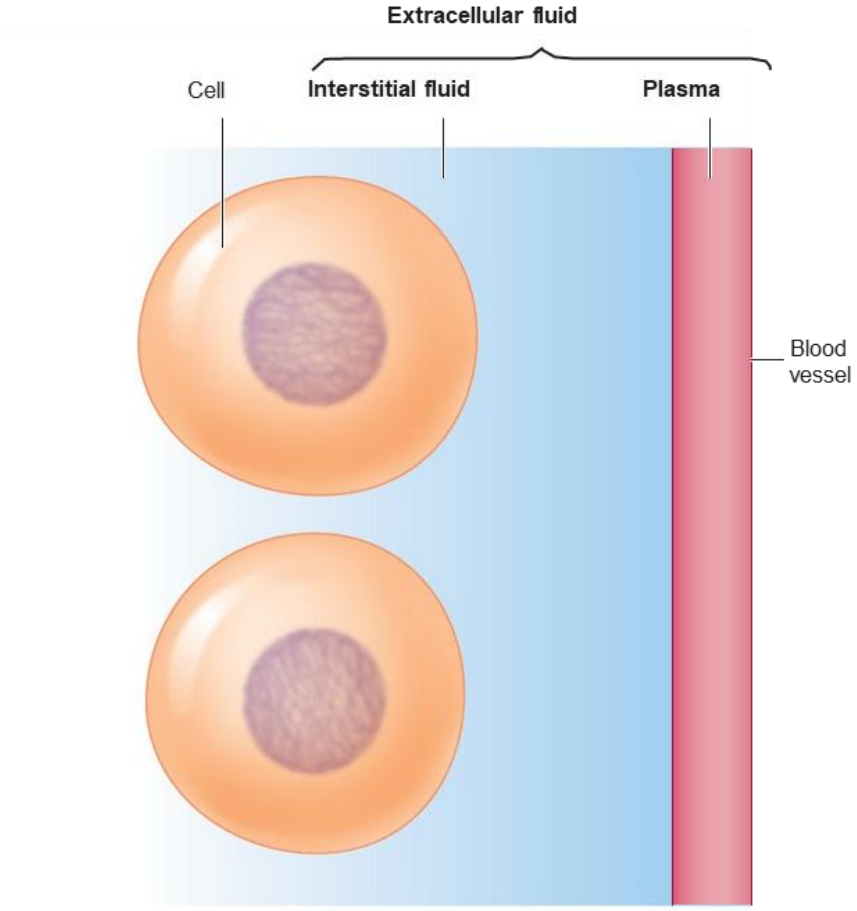
- External environment: surrounding environment where organism lives
- Internal environment: the fluid that surrounds the cells and where cells make life-sustaining exchanges
- Body cells are in contact with a privately maintained internal environment.
  - Intracellular fluid (ICF): a fluid contained within all body cells
  - Extracellular fluid (ECF): fluid outside the cells



https://science.mtchs.org/OnlineBio/BIOCD/text/chapter27/concept27.3.html

Components of the extracellular fluid

- Extracellular fluid (ECF): fluid outside the cells
  - Interstitial fluid, which surrounds, nourishes, washes the cells
  - Plasma, the fluid portion of blood



What are homeostatically regulated?

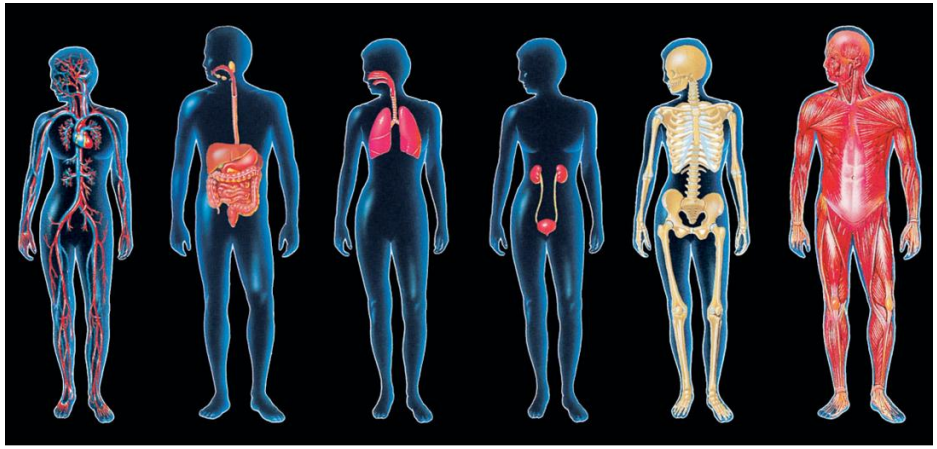
- Concentration of nutrients
- Concentration of O_2 and CO_2
- Concentration of waste products
- Changes in pH
- Concentrations of water, salt, and other electrolytes
- Volume and pressure
- Temperature

The body system - revisited

Body system is collection of organs performing related functions.

- Organs interact to accomplish a common activity essential for survival

11 body systems



Circulatory system
heart, blood vessels, blood

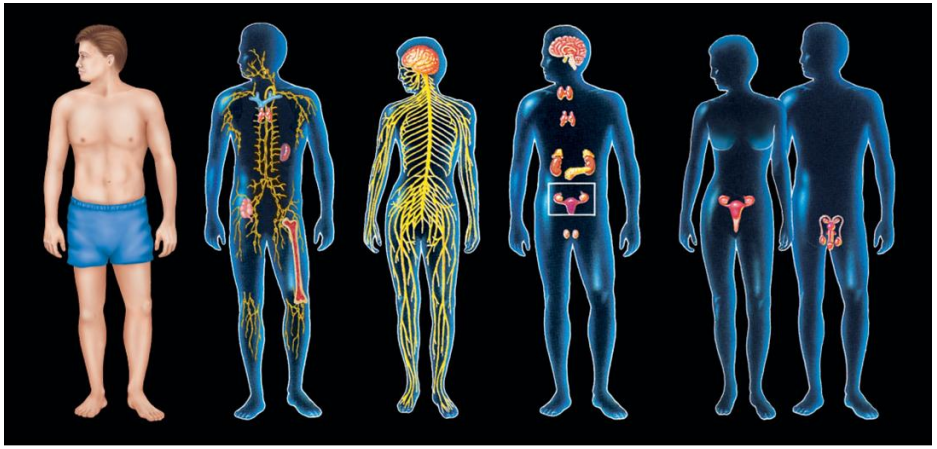
Digestive system
mouth, pharynx, esophagus, stomach, small intestine, large intestine, salivary glands, exocrine pancreas, liver, gallbladder

Respiratory system
nose, pharynx, larynx, trachea, bronchi, lungs

Urinary system
kidneys, ureters, urinary bladder, urethra

Skeletal system
bones, cartilage, joints

Muscular system
skeletal muscles



Integumentary system
skin, hair, nails

Immune system
lymph nodes, thymus, bone marrow, tonsils, adenoids, spleen, appendix, and, not shown, white blood cells, gut-associated lymphoid tissue, skin-associated lymphoid tissue

Nervous system
brain, spinal cord, peripheral nerves, and, not shown, special sense organs

Endocrine system
all hormone-secreting tissues, including hypothalamus, pituitary, thyroid, adrenals, endocrine pancreas, gonads, kidneys, pineal, thymus, and, not shown, parathyroids, intestine, heart, skin, adipose tissue

Reproductive system
Male: testes, penis, prostate gland, seminal vesicles, bulbourethral glands, associated ducts

Female: ovaries, oviducts, uterus, vagina, breasts

Each body systems contribute to homeostasis

- Circulatory system (heart, blood vessels, and blood): Transports materials from one part of the body to another
- Digestive system (mouth, esophagus, stomach, intestines, and related organs): Breaks down food into small nutrient molecules that can be absorbed
- Respiratory system (lungs and major airways): Gets O_2 from and eliminates CO_2 to the external environment
- Urinary system (kidneys and associated structures): Removes excess water, salt, acid, and other electrolytes from the plasma and eliminates them in the urine

Each body systems contribute to homeostasis

- Skeletal system (bones and joints): Provides support and protection for soft tissues and organs, Serves as a storage reservoir for calcium, Enables the body and its parts to move, Bone marrow is the source of all blood cells

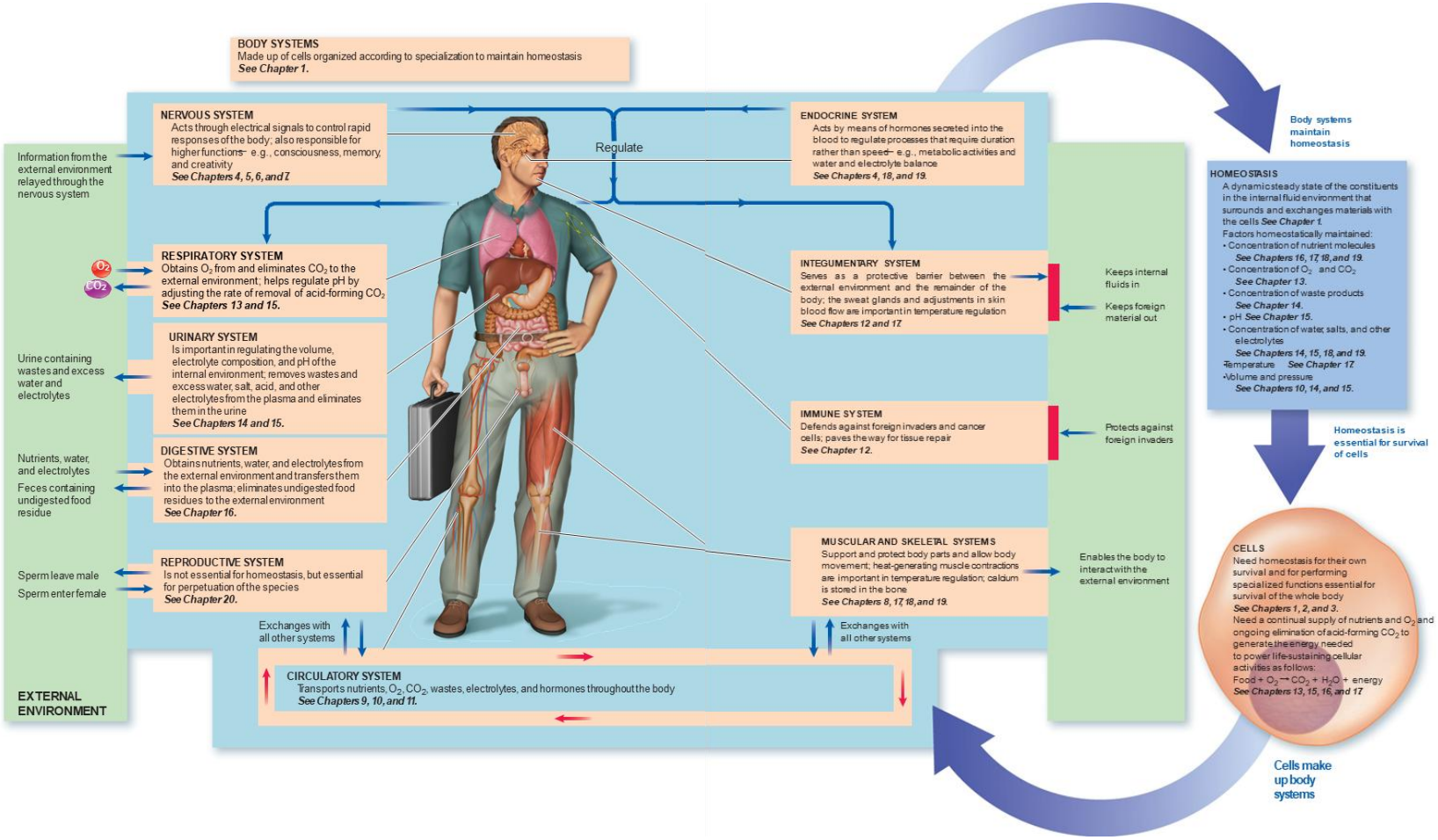
- Muscular system (skeletal muscles): Moves bones attached to skeletal muscles, Voluntary movements range from fine motor skills to powerful movements, Generates heat and maintains body temperature

- Integumentary system (skin and related structures): Serves as an outer protective barrier, Important in regulating body temperature

Each body systems contribute to homeostasis

- Immune system (white blood cells and lymphoid organs): Defends against foreign invaders such as bacteria and viruses and against body cells that have become cancerous, Helps in replacing injured or worn-out cells
- Nervous system (brain, spinal cord, nerves, and sense organs): Controls and coordinates body activities that require swift responses
- Endocrine system (all hormone-secreting glands): Regulates activities that require duration rather than speed, such as growth
- Reproductive system (male and female gonads and related organs): Essential for perpetuating the species, Not essential for homeostasis

Body systems maintain homeostasis (course summary)



1.4 Homeostatic control systems

Homeostatic control systems

- **What is a homeostatic control system?**

  - An interconnected network of body components
  - Operates to maintain a given factor in the internal environment at a relatively constant optimal level

  - To control homeostasis, the system must
    - Detect deviations from normal
    - Integrate this information with other information
    - Make adjustments to restore the factor to normal value

  - Homeostatic control systems operate through
    - Intrinsic control: locally controls inside an organ
    - Extrinsic control: regulatory mechanism initiated outside an organ to alter the activity of organ; accomplished by nervous or endocrine system

Negative and positive feedback, and feedforward

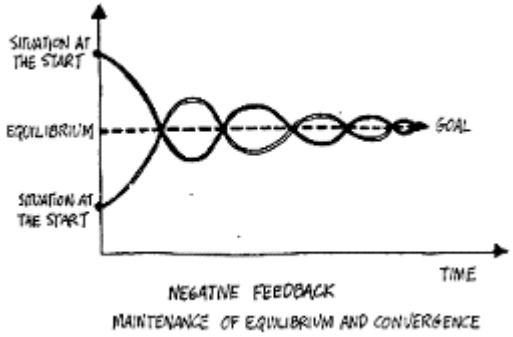
- Negative feedback opposes an initial change and is widely used to maintain homeostasis
- Change in a controlled variable triggers a response that drives the variable in the opposite direction of the initial change, thus opposing the change
- Example: controlling room temperature
  - the activity of organ; mainly accomplished by nervous or endocrine system)

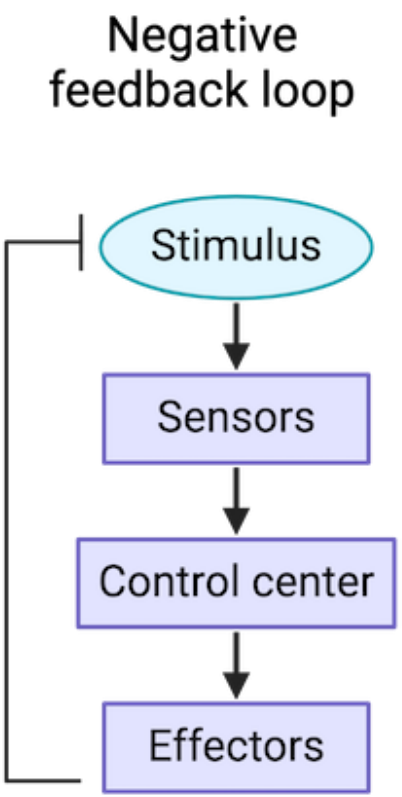
- In positive feedback, output enhances or amplifies a change so that the controlled variable continues to move in the direction of the initial change
- Less common than negative feedback but is important in some instances

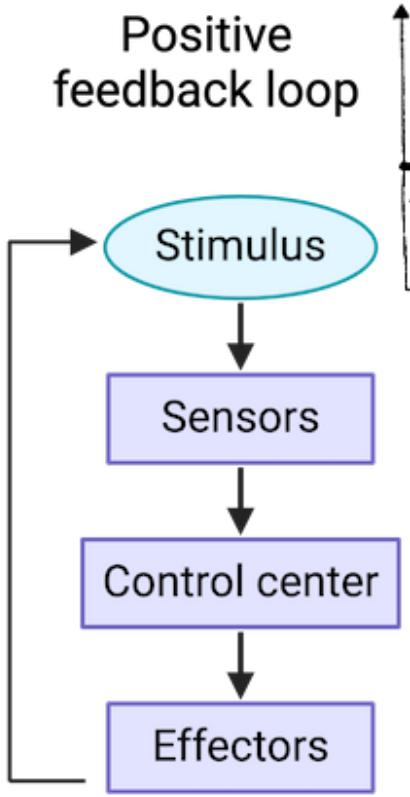
- Example: childbirth

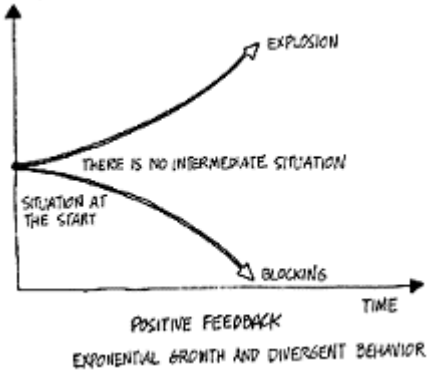
- Feedforward mechanisms initiate responses in anticipation of a change
  - ex) prepare to digest before eating, increase energy production before running

Definition of biopharmaceuticals



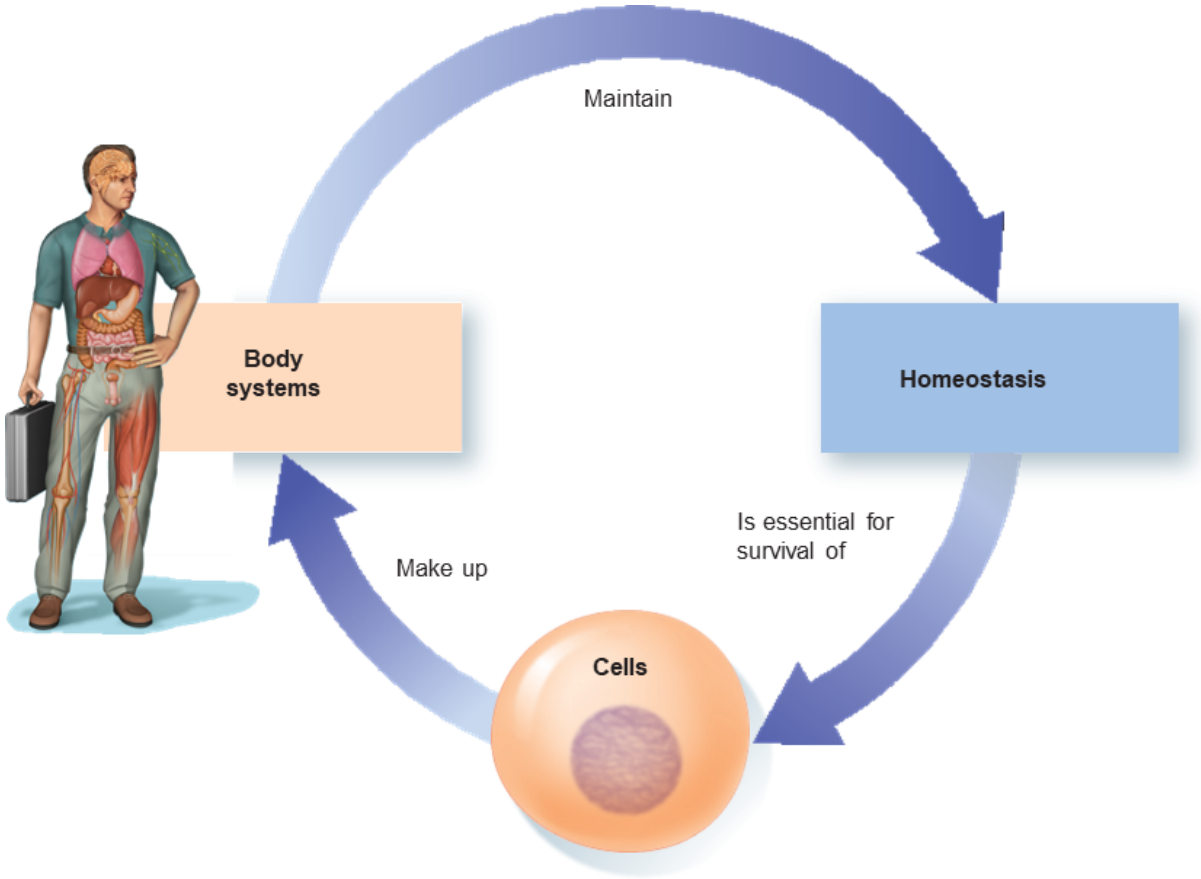






https://www.khanacademy.org/science/hs-bio/x230b3ff252126bb6:from-cells-to-organisms/x230b3ff252126bb6:homeostasis/a/homeostasis-and-feedback-loops
https://pespmc1.vub.ac.be/FEEDBACK.html

Concept of homeostasis



Thank you!

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Office hour: Wed 16-18 pm and anytime!